Experiment-7

Integration of Library Book Issue Form with Zoho CRM Using Connections and APIs

Aim:

To integrate the Library Book Issue application in Zoho Creator with Zoho CRM using authenticated Connections and APIs to transfer approved library-related data, maintain synchronization status, and prevent duplicate CRM records.

Overview:

Zoho Creator is a low-code application development platform used to create forms, reports, dashboards, and workflows. Zoho CRM is a customer relationship management platform that stores and manages contact information.

The integration connects these two platforms through an authenticated Connection and the Zoho CRM API.

In this experiment, the Library Book Issue form contains Student ID, Student Name, Department, Book ID, Book Title, Book Category, Issue Date, Due Date, and Issue Status.

The integration can be designed using the following components:

- Zoho Creator: Collects and stores library book issue details.
- Connection: Provides secure authorization for communication with Zoho CRM.
- API: Transfers the selected data from Creator to CRM.
- Field Mapping: Maps Creator fields to corresponding CRM fields.
- Workflow: Triggers the integration after a record is saved.
- Error Handling: Records failed synchronization attempts and helps prevent duplicate records.
- For this experiment, we can send selected student contact information to the CRM Contacts module. Library circulation details, such as book issue history and due dates, should remain in Creator unless there is an approved requirement to transfer them.

Procedure:

Step 1: Open the application

1. Log in to Zoho Creator.
2. Open the existing Library Book Issue application.
3. Verify that the form contains Student ID, Student Name, Department, and the other book issue fields.

Step 2: Configure the Zoho CRM Connection

1. Open the Connections section in Zoho Creator.
2. Select Zoho CRM as the service.
3. Create a Connection named `library_crm`.
4. Authorize access using the appropriate Zoho account.

5. Grant only the permissions required to create or update CRM contacts.

Step 3: Map the fields

Use the following mapping for the integration:

Zoho Creator field	Zoho CRM field
Student Name	Last Name
Student ID	External student identifier, if configured
Department	Department, if available in the selected CRM module

For a simplified experiment, Student Name can be mapped to the CRM `Last_Name` field. However, Zoho CRM requires a last name for Contacts, so a separate first-name and last-name mapping is preferable for a production application.

Note: Your current Creator form does not include an Email field. If you want to use email to identify existing CRM contacts and prevent duplicates, add an Email field and make it mandatory. Otherwise, configure an approved unique student identifier for matching.

Step 4: Create the integration workflow

1. Open Workflows in Zoho Creator.
2. Create a workflow for the Library Book Issue form.
3. Configure it to run after a record is successfully created.
4. Add a custom Deluge function or an integration task to send the selected information to Zoho CRM.

Step 5: Send data using the CRM API

The following is an illustrative Deluge request-building example based on the PDF's API-upsert method. It assumes you have added an Email field to your form and configured the `library_crm` Connection.

```
contact = Map();
contact.put("Last_Name", input.Student_Name);
contact.put("Email", input.Email);

items = List();
items.add(contact);

matchFields = List();
matchFields.add("Email");

payload = Map();
payload.put("data", items);
payload.put("duplicate_check_fields", matchFields);

response = invokeurl
[
    url: "https://www.zohoapis.com/crm/v8/Contacts/upsert"
    type: POST
    parameters: payload.toString()
    headers: {"Content-Type":"application/json"}
    connection: "library_crm"
];

info response;
```

Replace `Student_Name` and `Email` with the actual field link names shown in your Creator form. The API domain in this example is for the US data center; use the appropriate domain for your Zoho account.

This code demonstrates the request structure. Before using it, verify the current API requirements and add response validation and error handling. Do not treat the request as successful just because it returns a response.

Step 6: Handle the API response

1. Inspect the HTTP response and the individual CRM record result.
2. If the CRM reports success, store the returned CRM contact ID if your Creator form has a suitable field.
3. If the request fails, record a sanitized error message or code.

4. Retry temporary failures with a limited retry policy.
5. Do not retry authorization or validation errors endlessly.

Step 7: Test the integration

Test the following scenarios:

Test case	Expected result
Valid student contact	Contact is created or updated in CRM
Repeat the same request	Existing contact is matched; no duplicate is created
Missing required field	Clear validation error
Invalid Connection permissions	Authorization error is recorded
Temporary API failure	Failure is visible and can be retried safely

Step 8: Verify the CRM record

1. Open Zoho CRM.
2. Navigate to the Contacts module.
3. Search for the test student contact.
4. Verify the mapped fields.
5. Confirm that repeating the same request does not create a duplicate.

Result:

Thus, the integration approach between Zoho Creator and Zoho CRM was studied using authenticated Connections, API-based record upsert, field mapping, and error-handling procedures. The integration is

considered successful when a test contact is created or updated in CRM, duplicate requests are handled correctly, and failures are reported without incorrectly marking synchronization as successful.